Lab #2: Silver Equilibrium

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Abstract

In this experiment, the equilibrium of silver ion with chloride ion, iodide, and ammonia were studied by monitoring the time dependent electric potential curve of a electrochemical cell with Cu and Ag as electrodes. 1.00×10^{-4} M $AgNO_3$ solution was used to determine the cell constant. With potential stabilized, NH_4Cl, three portions of NH_3, and KI solution were added to determine silver concentrations and the Nernst equation. Debye-Huckel corrections were used to obtain solubility products for AgCl and AgI. A linear fit of $\log([\mathrm{Ag(NH_3)_p^+}]/[\mathrm{Ag^+}])$ versus $\log[NH_3]$ was used to estimate the coordination number and formation constant of the silver-ammonia complex. The calculated properties from experimental data were $K_{sp}(\mathrm{AgCl}) = (1.375 \pm 0.1050) \times 10^{-10}$ with an error rate of 22.32 % , $K_{sp}(\mathrm{AgI}) = (6.872 \pm 0.2277) \times 10^{-17}$ with an error rate of 19.34 % , and $K_f = (3.956 \pm 0.4929) \times 10^7$ with 95% confidence and an error rate of 147.3 % . The linear regression confirmed $p = 2.104 \pm 0.04850$, indicating that the dominant complex was $[\mathrm{Ag(NH_3)_2}]^+$.

Introduction

In a multi-component solution, different species of ions interact complicatedly with many physical and chemical processes happening, such as coordination, precipitation, etc. Determination of related property constants can help us predict the their chemical behavior and rationally design reactions.

The purpose of this experiment was to determine the equilibrium behavior of silver ions with three ligands, Cl^-, I^-, and NH_3 by measuring the time-dependent electric potential in a electrochemical cell. A silver and a copper electrodes are used to monitor changes in Ag^+ activity during a sequence of ligand additions.The observed potentials can be converted into free Ag^+ concentrations using the Nernst equation.[1]

$$E_{\mathrm{obs}} = C + 0.0592 \log_{10}[\mathrm{Ag^+}] \tag{1}$$

assuming the temperature is 25^oC. Once C is obtained from a solution of known silver concentration, later equilibrated electric potential can be used to compute $[\mathrm{Ag^+}]$. The silver ion concentrations can then be used to evaluate the solubility products of AgCl and AgI, the stoichiometry, and formation constant of the silver-ammonia complex.

Assume ideal conditions, the measured concentrations were used to evaluate[1]

$$Q_{sp}(AgX) = [\mathrm{Ag^+}][\mathrm{X^-}], \qquad K_{sp}(AgX) = \gamma_\pm^2 Q_{sp} \tag{2}$$

with $X^- = Cl^-$ or I^-. The activity coefficient $\gamma_\pm$ can be estimated from the Debye-Huckel law from[1]

$$\log \gamma_\pm = -|z_+ z_-| A I^{\frac{1}{2}} \tag{3}$$

where z is the number charges of ions in the compound and the ionic strength I is defined as $\frac{1}{2}\sum_i z_i^2 m_i$.

For the ammonia complex, the formation constant is[1]

$$K_f = \frac{[\mathrm{Ag(NH_3)_p^+}]}{[\mathrm{Ag^+}][\mathrm{NH_3}]^p}. \tag{4}$$

Assuming that essentially all dissolved silver is present as complex and that binding consumes a negligible fraction of the large excess of NH_3, the data can be rearranged into the linear form[1]

$$Y = \log\left(\frac{[\mathrm{Ag(NH_3)_p^+}]}{[\mathrm{Ag^+}]}\right) = \log K_f + p\log[\mathrm{NH_3}], \tag{5}$$

so that the slope gives the coordination number p and the intercept gives $\log K_f$. The properties of the ammonia complex can be computed and analyzed then.

Experimental

The experiment was carried out with a silver electrode and a copper electrode connected to a Logger Pro voltage interface. The voltage interface was calibrated with 1.5V dry cell and voltmeter before doing the experiment. Both electrolytes were placed in a glass electrochemical vessel. The silver electrode was first cleaned by immersion into hot sodium bicarbonate solution with aluminum foil before use. The copper electrode was placed in 0.1 M $CuSO_4$ solution, closely above a salt bridge prepared in advance. 10.00 mL of standardized 0.0010 M $AgNO_3$ and 0.010 M NH_4NO_3 was diluted to 100.0 mL using volumetric flask, and transferred to the electrochemical cell. Magnetic stir was used to equilibrate the solution during the whole experiment and get a stable baseline potential to get start with.

Voltage was then recorded every two seconds for a total length of 3000 seconds. First, 10.00 mL of 0.0010 M NH_4Cl was added to precipitate AgCl. Light white precipitation was observed shortly after adding the solution. After the potential stabilized, 10.00 mL mixture of 2.0 M NH_3 containing 0.010 M NH_4NO_3 were added three times, each followed by equilibration to a new potential plateau in around five minutes. Finally, 10.00 mL of 0.010 M KI was added and the final stabilized potential was recorded till the end. The solution became dark green first and turned milky afterwards. The potential-time curve was then analyzed in Logger Pro and their mean voltages and standard deviations were obtained.

The linear regression for Ag-NH_3 complex was plotted using Numpy[2], Matplotlib[3], and Pandas[4] libraries in Python. The linear regression parameters were determined by SciPy[5] library automatically.

Results

Figure 1 shows the potential-time curve with linear regressions done by Logger Pro. Six stable plateaus were identified and analyzed to obtain the silver concentrations at each stage. The mean potential and standard deviation of each plateau are collected in Table 1.

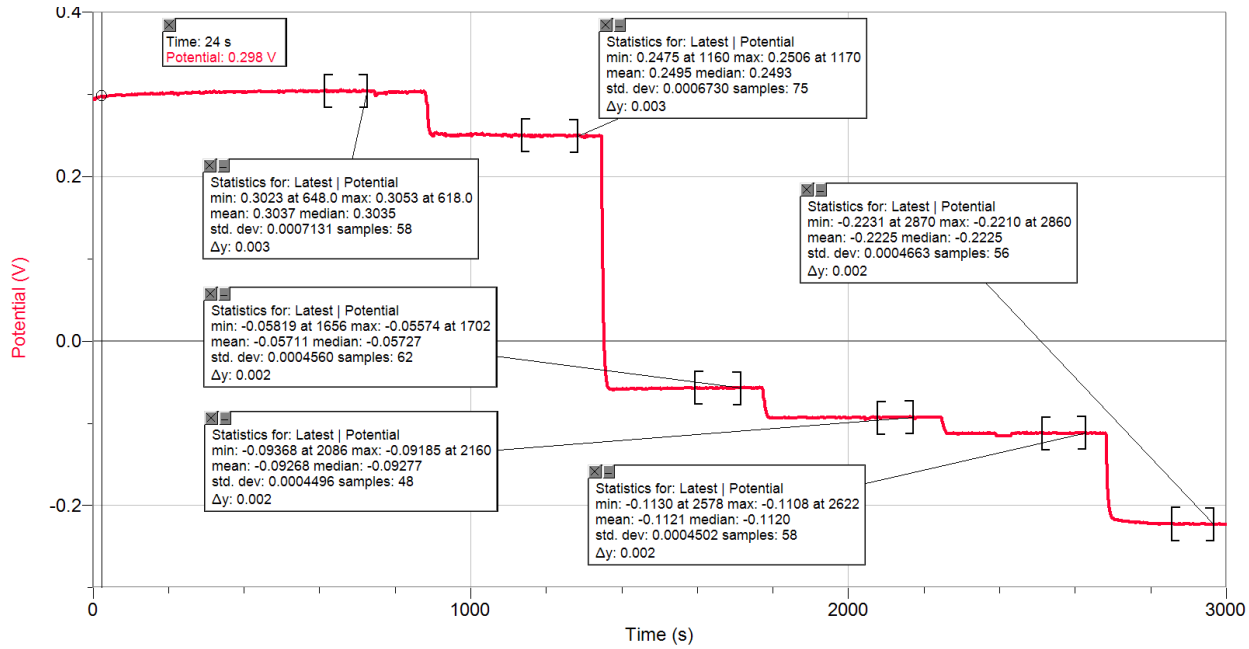

Figure 1: Voltage as a function of time during sequential addition of NH_4Cl, three aliquots of NH_3, and KI. The highlighted plateaus were used to calculate average stabilized potentials and their standard deviations.

Table 1: Stabilized cell potentials read from the processed Logger Pro trace.

Stage	E_{obs} / V	Standard deviation / V
Initial $AgNO_3$ solution (E_1)	0.3037	0.0007131
After NH_4Cl addition (E_2)	0.2495	0.0006730
After 10 mL NH_3 (E_3)	-0.05711	0.0004560
After 20 mL NH_3 (E_4)	-0.09268	0.0004496
After 30 mL NH_3 (E_5)	-0.1121	0.0004502
After KI addition (E_6)	-0.2225	0.0004663

Nerst Constant

Using Eq.1 and $[Ag^+] = 1.00 \times 10^{-4} M$ for the initial solution,

$$C = E_0 - 0.0592 \log(1.00 \times 10^{-4}) = 0.5405 \pm 0.0007131 \text{ V}. \tag{6}$$

where the uncertainty in C was taken directly as the standard deviation of the potential with initial Ag^+ concentration was treated as exact.

AgCl Equilibrium

After addition of NH_4Cl, the silver concentration was obtained from Eq.1

$$[Ag^+]_{\text{AgCl}} = 10^{(E_{\text{AgCl}} - C)/0.0592} = (1.2147 \pm 0.0463) \times 10^{-5} \text{ M}. \tag{7}$$

As the initial Ag^+ and Cl^- concentrations were equal before precipitation, we can assume that $[Ag^+] = [Cl^-]$, giving

$$Q_{sp}(\text{AgCl}) = [\text{Ag}^+]^2 = (1.475 \pm 0.113) \times 10^{-10}. \tag{8}$$

After dilution to 110 mL, the concentration of NH_4NO_3 was 9.09×10^{-4} M, so $I = 9.09 \times 10^{-4}$ and $\gamma_\pm = 0.9653$. Therefore,

$$K_{sp}(\text{AgCl}) = \gamma_\pm^2 Q_{sp} = (1.375 \pm 0.1050) \times 10^{-10}. \tag{9}$$

Silver-ammonia Complex

The silver concentrations after the three ammonia additions can be calculated as

$$[\text{Ag}^+]_{E_3} = (8.04 \pm 0.26) \times 10^{-11} \text{ M} \tag{10}$$
$$[\text{Ag}^+]_{E_4} = (2.016 \pm 0.066) \times 10^{-11} \text{ M} \tag{11}$$
$$[\text{Ag}^+]_{E_5} = (9.47 \pm 0.31) \times 10^{-12} \text{ M}. \tag{12}$$

The concentrations of NH_3 and $Ag(NH_3)_p^+$ in diluted solution are calculated as

$$[\text{Ag}(\text{NH}_3)_\text{p}^+] = (1.000 \times 10^{-4}) \frac{100}{110 + V} \tag{13}$$

and

$$[\text{NH}_3] = \frac{V(2.00)}{110 + V}, \tag{14}$$

values of Y and $\log[NH_3]$ were calculated for $V = 10$, 20, and 30 mL.

Linear regression of Eq.5 show that

$$Y = (2.104 \pm 0.04850) \log[NH_3] + (7.655 \pm 0.02805), \qquad R^2 = 0.9999. \tag{15}$$

The regression is shown in Figure 2. The slope corresponds to $p = 2.104 \pm 0.04850$, which rounds to $p = 2$.

As $logK_f = b$, from the intercept, the graph-based formation constant is

$$K_{f,\text{graph}} = 10^{7.655} = (4.515 \pm 0.2911) \times 10^7. \tag{16}$$

Using the rounded integer value $p = 2$, the formation constants are calculated from the three individual points

$$K_{f,1} = 3.732 \times 10^7 \tag{17}$$
$$K_{f,2} = 4.031 \times 10^7 \tag{18}$$
$$K_{f,3} = 4.107 \times 10^7. \tag{19}$$

With averaging, error propagation and analysis, we can have

$$\bar{K}_f = 3.956 \times 10^7, \qquad S_{K_f} = 1.984 \times 10^6. \tag{20}$$

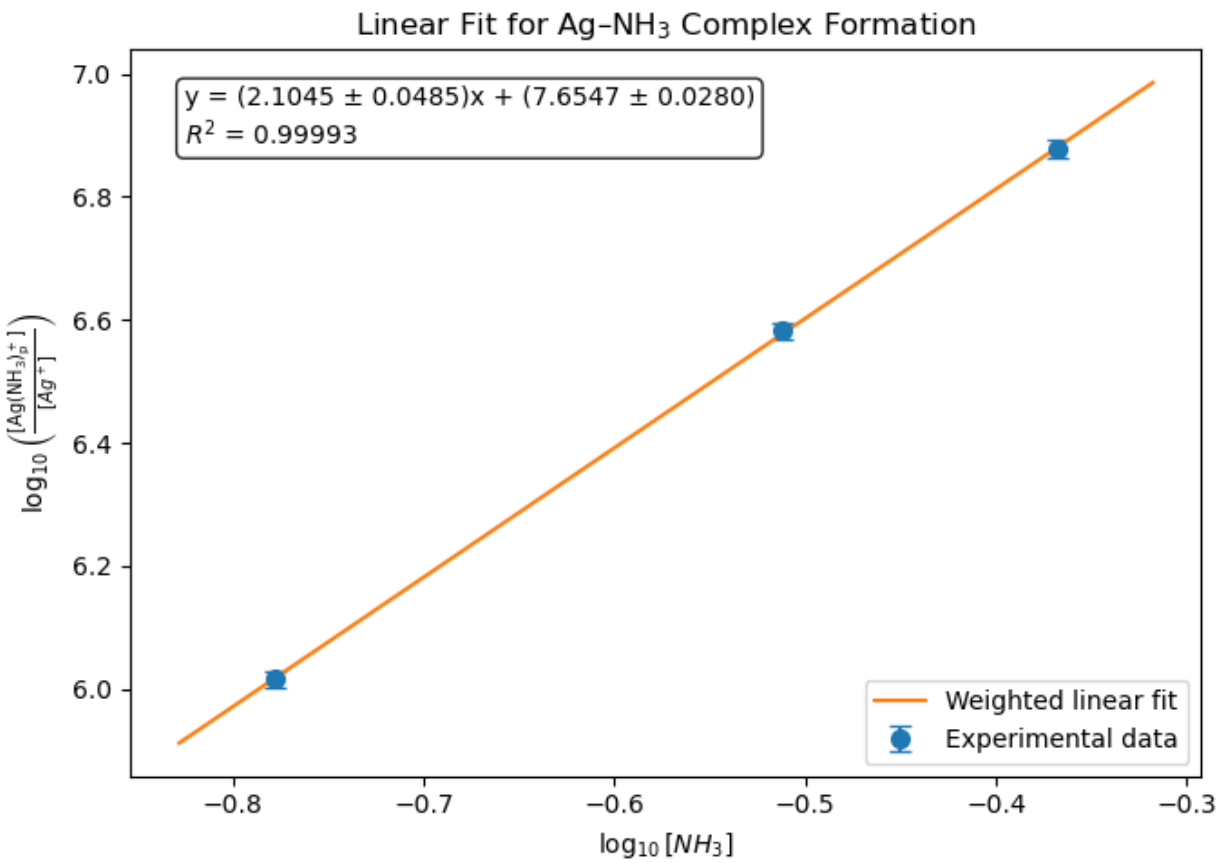

Figure 2: Linear regression of $Y = \log([\mathrm{Ag(NH_3)_p^+}]/[\mathrm{Ag^+}])$ versus $\log[NH_3]$. The displayed fit reports the slope ($p = 2.10 \pm 0.05$) and intercept ($b = 7.65 \pm 0.03$), with $R^2 = 0.9999$.

Using $t = 4.303$ for $n = 3$ at 95% confidence,

$$\lambda_{K_f} = t\frac{S_{K_f}}{\sqrt{n}} = 4.93 \times 10^6, \tag{21}$$

so the final pointwise estimate is

$$K_f = (3.956 \pm 0.4929) \times 10^7 \tag{22}$$

with 95% confidence.

AgI equilibrium

The final stabilized electric potential gives that

$$[\mathrm{Ag^+}]_{\mathrm{AgI}} = (1.293 \pm 0.043) \times 10^{-13}\ \mathrm{M}. \tag{23}$$

As suggested in the guideline, because KI was added in largely excess relative to the initial silver concentration, the iodide concentration could not be approximated as equal to the

silver concentration. Instead, the remaining iodide was calculated from the initial moles of iodide minus the moles required to precipitate the initial silver concentrations.

$$[\mathrm{I}^-]_{\mathrm{left}} = 6.00 \times 10^{-4}\ \mathrm{M}. \tag{24}$$

Thus,

$$Q_{sp}(\mathrm{AgI}) = [\mathrm{Ag}^+][\mathrm{I}^-]_{\mathrm{left}} = (7.756 \pm 0.2570) \times 10^{-17}. \tag{25}$$

At this stage the NH_4NO_3 concentration was 2.66710^{-3} M, given $\gamma_\pm = 0.9413$.

$$K_{sp}(\mathrm{AgI}) = \gamma_\pm^2 Q_{sp} = (6.872 \pm 0.2277) \times 10^{-17}. \tag{26}$$

The final numerical results are summarized in Table 2.

Table 2: Summary of calculated equilibrium constants. Literature values for K_{sp} were taken from the CRC Handbook[6], and the literature value for K_f was taken from the LibreTexts table[7] of complex-ion formation constants.

Quantity	Experimental value	Literature value	Error Rate
K_{sp}(AgCl)	$(1.375 \pm 0.1050) \times 10^{-10}$	1.77×10^{-10}	22.32 %
K_{sp}(AgI)	$(6.872 \pm 0.2277) \times 10^{-17}$	8.52×10^{-17}	19.34 %
K_f from graph	$(4.515 \pm 0.2911) \times 10^{7}$	1.6×10^{7}	182.2 %
Averaged K_f	$(3.956 \pm 0.4929) \times 10^{7}$	1.6×10^{7}	147.3 %

Discussion

In general, the electrochemical method basically described three distinct kinds of silver equilibrium with voltage-time curve. The AgCl and AgI plateaus were well separated from the baseline potential, and the ammonia additions gradually decreased the free silver concentration. The linearity of the Y-log$[NH_3]$ plot was great ($R^2 = 0.9999$), and the fitted slope $p = 2.104 \pm 0.04850$ shows that the dominant dissolved complex was $[\mathrm{Ag(NH_3)_2}]^+$. This conclusion is chemically reasonable because silver(I) ion commonly forms a linear diammine complex in aqueous ammonia solutions.

The measured K_{sp}(AgCl) was lower than the literature value by 22.32%, while K_{sp}(AgI) was lower by 19.34%. These two deviations are similar in magnitude and direction, which suggests the presence of a common systematic contribution. The most likely source is the use of 25^oC in the Nernst slope rather than the actual measured temperature. A small temperature mismatch affects all converted Ag^+ concentrations in the same direction and introduce errors in related properties. Another systematic error could arise from the assumption that NH_4NO_3 alone determines the ionic strength for both salts. Although this approximation largely simplifies calculations, it neglects small contributions from the reacting ions themselves.

The formation constant showed a larger discrepancy from the literature value, where the graph-based value is higher the literature value by 182%, and the averaged value is

higher by 147.3%. It's shown that the pointwise averaging method is more accurate. The complex-formation calculation is particularly sensitive because $[Ag^+]$ after ammonia addition is extremely small. So, a small uncertainty in electric potential can propagate into a large fractional uncertainty in K_f. In addition, the intercept method determines K_f by extrapolating the regression to $\log[NH_3] = 0$, which lies outside the narrow range of measured ammonia concentrations. Also, the initial calibration of the Logger Pro voltmeter could also systematically alter the results, resonating with other errors in K_{sp}.

Overall, the experiment demonstrated that emf measurements can be used to determine silver equilibrium constants in dilute solution. The main conclusions are that the dominant ammonia complex was $[Ag(NH_3)_2]^+$, the measured solubility products were of the correct order of magnitude but somewhat lower than literature values, and the formation constant obtained from the present dataset was systematically higher than the accepted value. These errors can be further corrected by improving experimental procedures and apparatus.

Conclusion

The potential measurements obtained with the silver electrode were used to determine the solubility products of AgCl and AgI and to analyze the formation of the silver-ammonia complex. From the AgCl portion of the experiment, the activity-corrected solubility product was found to be $K_{sp}(\text{AgCl}) = (1.375 \pm 0.1050) \times 10^{-10}$. From the AgI portion, the corrected value was $K_{sp}(\text{AgI}) = (6.872 \pm 0.2277) \times 10^{-17}$. Linear regression of $\log([\text{Ag}(\text{NH}_3)_\text{p}^+]/[Ag^+])$-$\log[NH_3]$ gave $p = 2.104 \pm 0.04850$, indicating that the dominant complex was $[\text{Ag}(\text{NH}_3)_2]^+$. The corresponding K_f were calculated to be $4.515 \pm 0.2911 \times 10^7$ directly from the graph and $3.956 \pm 0.4929 \times 10^7$ from averaging points.

Overall, the experiment verified the Nernst equation and activity corrections in equilibrium analysis. The experimental constants deviated from literature values, likely due to voltameter calibration and the approximations used in calculations.

References

[1] Henry C. Brenner, D. S. J. W. A. H. N. K. N. E. N. M. K. P. S. C. S., Pance Naumov *Laboratory Manual of Physical Chemistry*; 2015; Experiment 6: Silver Equilibrium.

[2] Harris, C. R. et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362.

[3] Hunter, J. D. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering* **2007**, *9*, 90–95.

[4] pandas development team, T. pandas-dev/pandas: Pandas. 2020; `https://doi.org/10.5281/zenodo.3509134`.

[5] Virtanen, P. et al. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* **2020**, *17*, 261–272.

[6] Lide, D. Handbook of Chemistry and Physics; CRC Press: Boca Raton, FL. **2002**,

[7] University of California, Davis Table of formation constants of complex ions. LibreTexts-libraries.

Appendix

The raw data and corresponding analysis script are available on GitHub

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1. Determine the Nerst Equation

From raw data:

$E_1 = 0.3037 \pm 0.0007131\ V$

$E_2 = 0.2495 \pm 0.0006730\ V$

$E_3 = -0.05711 \pm 0.0004560\ V$

$E_4 = -0.09268 \pm 0.0004496\ V$

$E_5 = -0.1121 \pm 0.0004502\ V$

$E_6 = -0.2225 \pm 0.0004663\ V$

From guide, the Nerst equation is given by:

$$E_{obs} = const. + 0.0592 \lg[Ag^+]$$

As the initial concentration is $[Ag^+]_0 = 1.00\times10^{-4} M$,

$$\begin{aligned} Const. &= E_1 - 0.0592 \lg(1.00\times10^{-4}) \\ &= 0.3037 \pm 0.0007131\ V + 4\cdot 0.0592 \\ &= 0.5405 \pm 0.0007131\ V \end{aligned}$$

2. Calculation for AgCl

(1) Calculation of $[Ag^+]$ & Q_{sp}

Using Nerst equation:

$$E_{obs} = const. + 0.0592 \lg[Ag^+]$$

$$[Ag^+] = 10^{(E_{obs} - 0.5405)/0.0592}$$

Figure 3: Computations of Values and Error Analysis (Page 1)

Use E_2: $[Ag^+]_{AgCl} = 10^{(0.2495-0.5405)/0.0592}\ M$

$= 10^{-0.291/0.0592}\ M$

$\approx 1.215\times10^{-5}\ M$

Assume $[Ag^+]=[Cl^-]$,

$Q_{sp}(AgCl) = [Ag^+]^2_{AgCl} = (10^{-0.291/0.0592})^2 \approx 1.475\times10^{-10}$

(2) Error Analysis I

Let $y = 10^{(E_{obs}-C)/0.0592}$,

① $\frac{\partial y}{\partial E_{obs}} = 10^{(E_{obs}-C)/0.0592}\cdot \ln(10)\cdot\frac{1}{0.0592} = y\frac{\ln 10}{0.0592}$

② $\frac{\partial y}{\partial C} = 10^{(E_{obs}-C)/0.0592}\cdot \ln(10)\left(-\frac{1}{0.0592}\right) = -y\frac{\ln 10}{0.0592}$

$S_y = \sqrt{\left(y\frac{\ln 10}{0.0592}S_E\right)^2 + \left(-y\frac{\ln 10}{0.0592}S_C\right)^2} = y\frac{\ln 10}{0.0592}\sqrt{S_E^2+S_C^2}$

From previous calculations & experimental data:

$S_E = 0.0006730 \quad S_C = 0.0007131$

$S_y = 1.215\times10^{-5}\cdot\frac{\ln 10}{0.0592}\cdot\sqrt{(6.730\times10^{-4})^2+(7.131\times10^{-4})^2} \approx 0.0463\times10^{-5}$

Thus, $[Ag^+]_{AgCl} = (1.215\pm0.0463)\times10^{-5}\ M$

As $Q = c_{Ag^+}c_{Cl^-} = c^2_{Ag^+}$, $\frac{dQ}{dc} = 2c$

$S_Q = 2\cdot c\cdot S_c = 2\cdot 1.215\times10^{-5}\cdot 4.634\times10^{-7} \approx 0.113\times10^{-10}$

So, $Q(AgCl) = (1.475\pm0.113)\times10^{-10}$

Figure 4: Computations of Values and Error Analysis (Page 2)

(3) Calculation for $\gamma_\pm$

$n(NH_4NO_3) = 0.01M \cdot 0.010L = 1.00\times10^{-4} mol$

$V_{total} = 10mL + 100mL = 0.110L$

$[NH_4NO_3] = \frac{1.00\times10^{-4}}{0.110} M \approx 9.09\times10^{-4} M$

From lecture: $I = \frac{1}{2}\sum_i z_i^2 m_i$, $z_1 = z_2$ & $m_1 = m_2$ for NH_4NO_3

As $[NH_4^+] = [NO_3^-]$, assume 25°C, $A = 0.5092$, from lecture:

$$\log\gamma_\pm = -|z_+ z_-| A I^{\frac{1}{2}}$$
$$= -0.5092 I^{\frac{1}{2}}$$
$$= -0.5092 \cdot \sqrt{\frac{1}{2}\cdot(1+1)\cdot\frac{1.00\times10^{-4}}{0.110}}$$
$$\approx -0.01535$$
$$\gamma_\pm = 10^{-0.01535} \approx 0.9653$$

(4) Calculation for K_{sp}

$K_{sp} = \gamma_\pm^2 Q_{sp} = 0.9653^2 \cdot (1.475\times10^{-10}) \approx 1.375\times10^{-10}$

(5) Error Analysis II

Since $K = \gamma_\pm^2 Q$, $S_K = \gamma_\pm^2 S_Q = 0.9653^2 \cdot 0.113\times10^{-10} \approx 0.105\times10^{-10}$

So, $K_{sp}(AgCl) = (1.375 \pm 0.105)\times10^{-10} M$

Literature: $K_{sp} = 1.77\times10^{-10}$

error rate $= \frac{|(1.375-1.77)\times10^{-10}|}{1.77\times10^{-10}}\times100\% = \frac{0.395}{1.77}\times100\% \approx 22.32\%$

Figure 5: Computations of Values and Error Analysis (Page 3)

3. Calculation for AgI

(1) Calculation for $[Ag^+]$, $[I^-]$ & Q_{sp}

Use previously derived Nerst equation & E_6:

$$[Ag^+]_{AgI} = 10^{(-0.2225-0.5405)/0.0592} \approx 1.293\times10^{-13}\,M$$

$$n(I^-)_{total} = 0.010L \cdot 0.01M = 1.00\times10^{-4}\,mol$$

$$n(Ag^+)_{ini} = 0.010L \cdot 0.001M = 1.00\times10^{-5}\,mol$$

$$n(I^-)_{res} = 1.00\times10^{-4}mol - 1.00\times10^{-5}mol = 9.00\times10^{-5}mol$$

$$V_{final} = 100mL + 10mL + 30mL + 10mL = 0.150L$$

$$[I^-]_{res} = \frac{9.00\times10^{-5}}{0.150}M = 6.00\times10^{-4}M$$

$$Q_{sp} = [Ag^+]_{AgI}\cdot[I^-]_{res} = 10^{(-0.2225-0.5405)/0.0592}\cdot 6.00\times10^{-4} \approx 7.756\times10^{-13}$$

(2) Error Analysis I

From previous:

$$S(Ag^+) = [Ag^+]_{AgI}\frac{\ln10}{0.0592}\sqrt{S_{E_6}^2 + S_C^2}$$

$$= 10^{(-0.2225-0.5405)/0.0592}\cdot\frac{\ln10}{0.0592}\sqrt{(4.663\times10^{-4})^2 + (2.131\times10^{-4})^2}$$

$$\approx 0.043\times10^{-13}$$

So, $[Ag^+]_{AgI} = (1.293 \pm 0.043)\times10^{-13}\,M$

$$S_Q = [I^-]_{res}\cdot S(Ag^+) = 6.00\times10^{-4}\cdot 4.2838\times10^{-15} \approx 0.2570\times10^{-17}$$

So, $Q_{sp} = (7.756 \pm 0.2570)\times10^{-17}$

Figure 6: Computations of Values and Error Analysis (Page 4)

(3) Calculation for $\gamma_\pm$

$n_{total}(NH_4NO_3) = 1.00\times10^{-4}\,mol \cdot 4 = 4.00\times10^{-4}\,mol$

$[NH_4NO_3] = \frac{4.00\times10^{-4}}{0.150}\,M \approx 2.667\times10^{-3}\,M$

From previous, $\gamma_\pm = 10^{-0.509\cdot\sqrt{2.667\times10^{-3}}} \approx 0.9413$

(4) Calculation for K_{sp}

$K_{sp} = \gamma_\pm^2 Q_{sp} = (0.9413)^2 \cdot (7.259\times10^{-17}) \approx 6.872\times10^{-17}$

(5) Error Analysis II

$S_K = \gamma_\pm^2 \cdot S_Q = 0.9413^2 \cdot 2.570\times10^{-18} \approx 2.277\times10^{-18}$

So, $K_{sp} = (6.872 \pm 0.2277)\times10^{-17}$

Literature: $K_{sp}(AgI) = 8.52\times10^{-17}$

error rate $= \frac{|(8.52-6.872)\times10^{-17}|}{8.52\times10^{-17}} \times 100\% \approx 19.34\%$

4. Formation Constant & K_f

(1) $[Ag^+]$, $[NH_3]$ & $[Ag(NH_3)^+]$

$[Ag^+]_3 = 10^{(E_3-C)/0.0592} = 10^{(-0.2571-0.5405)/0.0592} \approx 8.040\times10^{-11}\,M$

$S_{[Ag^+]_3} = [Ag^+]_3 \frac{\ln 10}{0.0592}\sqrt{S_{E_3}^2+S_C^2} \approx 0.2647\times10^{-11}\,M$

$[NH_3]_3 = \frac{10\cdot2.00}{120}\,M \approx 0.1667\,M$

$[Ag(NH_3)^+]_3 = 1.000\times10^{-4}\cdot\frac{100}{120} \approx 8.333\times10^{-5}\,M$

So, $[Ag^+]_3 = (8.040 \pm 0.2647)\times10^{-11}\,M$

Figure 7: Computations of Values and Error Analysis (Page 5)

$$[Ag^+]_4 = 10^{(E_4 - C)/0.0592} = 10^{(-0.2268 - 0.5405)/0.0592} \approx 2.016 \times 10^{-11} M$$

$$S_{[Ag^+]_4} = [Ag^+]_4 \frac{\ln 10}{0.0593} \sqrt{S_{E_4}^2 + S_C^2} \approx 0.066 \times 10^{-11} M$$

$$[NH_3]_4 = \frac{20.0 \cdot 0.20}{130} M \approx 0.3077 M$$

$$[Ag(NH_3)^+]_4 = 1.000 \times 10^{-4} \cdot \frac{100}{130} \approx 7.692 \times 10^{-5} M$$

So $[Ag^+]_4 = (2.016 \pm 0.066) \times 10^{-11} M$

$$[Ag^+]_5 = 10^{(E_5 - C)/0.0592} = 10^{(-0.1121 - 0.5405)/0.0592} \approx 9.470 \times 10^{-12} M$$

$$S_{[Ag^+]_5} = [Ag^+]_5 \frac{\ln 10}{0.0593} \sqrt{S_{E_5}^2 + S_C^2} \approx 0.3106 \times 10^{-11} M$$

$$[NH_3]_5 = \frac{30.0 \cdot 0.20}{140} M \approx 0.4286 M$$

$$[Ag(NH_3)^+]_5 = 1.000 \times 10^{-4} \cdot \frac{100}{140} \approx 7.143 \times 10^{-5} M$$

So $[Ag^+]_5 = (9.470 \pm 0.3106) \times 10^{-11} M$

(2) Linear Regression

Point 1: $\log(NH_3) = \log(0.1667) \approx -0.7782$

$$\log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right) = \log\left(\frac{8.233 \times 10^{-5}}{8.040 \times 10^{-11}}\right) \approx 6.016$$

Let $y = \log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right)$, $\frac{\partial y}{\partial [Ag^+]} = -\frac{1}{[Ag^+] \ln 10}$

$$S_y = \frac{S_{[Ag^+]}}{[Ag^+] \ln 10} = \frac{2.647 \times 10^{-12}}{8.040 \times 10^{-11} \ln 10} \approx 0.01430$$

Point 2: $\log(NH_3) = \log(0.3077) \approx -0.5119$

$$\log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right) = \log\left(\frac{7.692 \times 10^{-5}}{2.016 \times 10^{-11}}\right) \approx 6.582$$

$$S_y = \frac{S_{[Ag^+]}}{[Ag^+] \ln 10} = \frac{0.066 \times 10^{-11}}{2.016 \times 10^{-11} \ln 10} \approx 0.01422$$

Figure 8: Computations of Values and Error Analysis (Page 6)

Point 3: $\log(NH_3) = \log(0.4286) \approx -0.3680$

$$\log\left(\frac{[Ag(NH_3)_2^+]}{[Ag^+]}\right) = \log\left(\frac{7.143\times10^{-5}}{9.470\times10^{-11}}\right) \approx 6.878$$

$$S_\gamma = \frac{S_{[Ag^+]}}{[Ag^+]\ln 10} = \frac{0.3106\times10^{-11}}{9.470\times10^{-11}\ln 10} \approx 0.01424$$

From linear regression,

$m = 2.1045 \quad S_m = 0.0485$

$b = 7.6547 \quad S_b = 0.0280$

So, $p = 2.104 \pm 0.04850$. Take $p = 2$

As $\log K_f = b$, $K_f = 10^{7.6547} \approx 4.515\times10^7$

Since $K_f = 10^b$, $\frac{dK_f}{db} = \ln 10 \cdot 10^b = \ln 10 K_f$

$$S_{K_f} = \ln 10 \cdot K_f \cdot S_b = \ln 10 \cdot 4.515\times10^7 \cdot 0.0280 \approx 2.911\times10^6$$

So, $K_f = (4.515 \pm 0.2911)\times10^7$

(3) Calculation for K_f from p

As $p=2$, $K_f = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2}$

Point 1: $K_{f_1} = \frac{8.333\times10^{-5}}{(8.040\times10^{-11})(0.1667)^2} \approx 3.732\times10^7$

$$S_{K_{f_1}} = K_f \frac{S_{[Ag^+]_3}}{[Ag^+]_3} = 3.732\times10^7 \cdot \frac{2.647\times10^{-12}}{8.040\times10^{-11}} \approx 1.229\times10^6$$

Point 2: $K_{f_2} = \frac{7.692\times10^{-5}}{(2.016\times10^{-11})(0.3077)^2} \approx 4.031\times10^7$

$$S_{K_{f_2}} = 4.031\times10^7 \cdot \frac{0.66\times10^{-12}}{2.016\times10^{-11}} \approx 1.328\times10^6$$

Figure 9: Computations of Values and Error Analysis (Page 7)

Point 3: $K_{f_3} = \frac{2.143 \times 10^{-5}}{(9.470 \times 10^{-11})(0.4286)^2} \approx 4.107 \times 10^7$

$$S_{K_{f_3}} = 4.107 \times 10^7 \cdot \frac{3.186 \times 10^{-12}}{9.470 \times 10^{-11}} \approx 1.347 \times 10^6$$

$$\overline{K_f} = \frac{K_{f_1} + K_{f_2} + K_{f_3}}{3} \approx 3.956 \times 10^7$$

$$S_{\overline{K_f}} = \sqrt{\frac{\sum_i (K_{f_i} - \overline{K_f})}{2}} \approx 1.984 \times 10^6$$

From table, $t = 4.303$ for 95% interval

$$\lambda_{K_f} = t \frac{S_{K_f}}{\sqrt{n}} = 4.303 \frac{1.984 \times 10^6}{\sqrt{3}} \approx 0.4929 \times 10^7$$

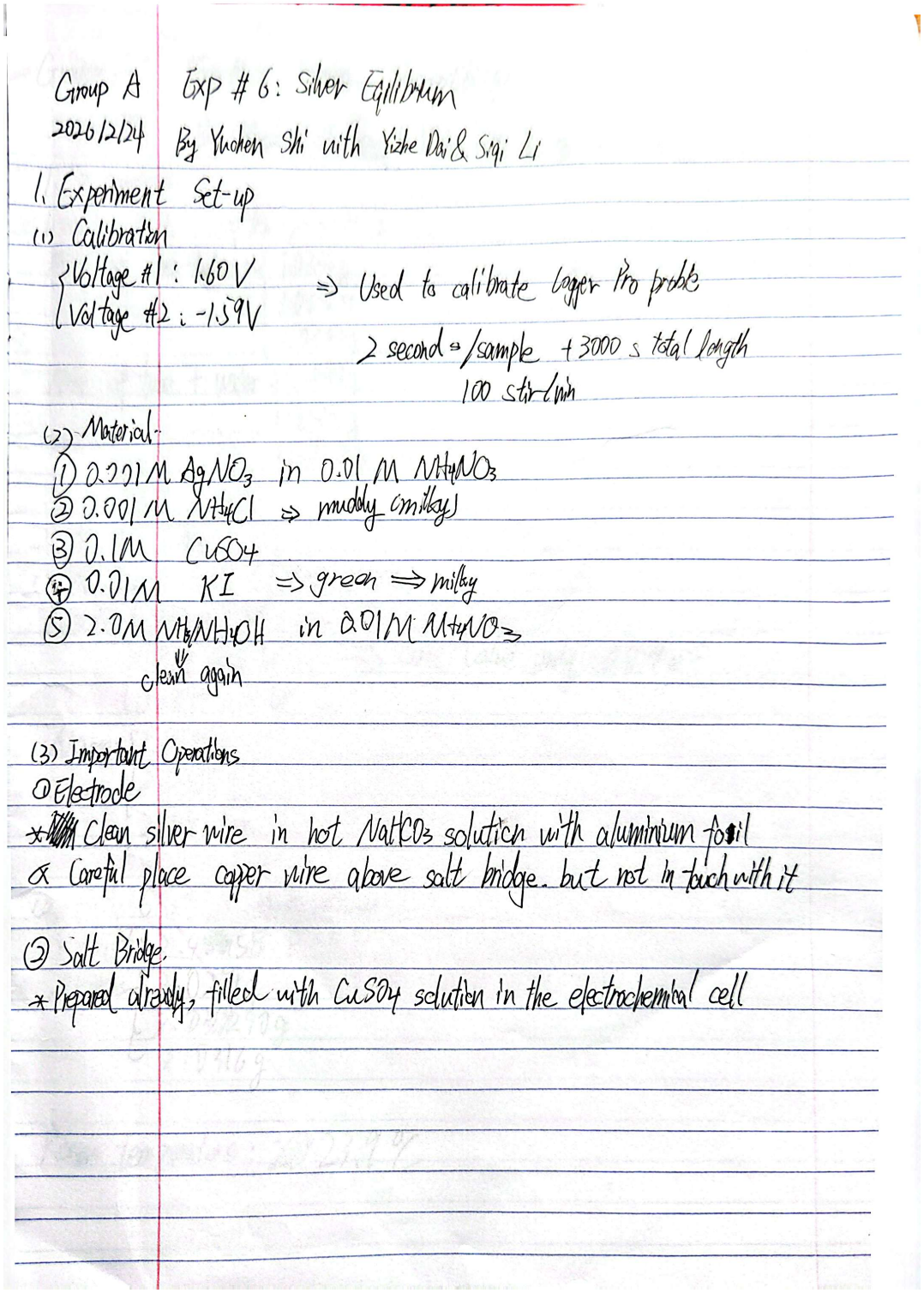
So, $K_f = (3.956 \pm 0.4929) \times 10^7$

From literature $K_f = 1.6 \times 10^7$,

error rate 1 $= \frac{|4.515 - 1.61| \times 10^7}{1.6 \times 10^7} \times 100\% \approx 182.2\%$

error rate 2 $= \frac{|3.956 - 1.6| \times 10^7}{1.6 \times 10^7} \times 100\% \approx 147.3\%$

Figure 10: Computations of Values and Error Analysis (Page 8)



Group A Exp # 6: Silver Equilibrium

2026/2/24 By Yuchen Shi with Yizhe Dai & Siqi Li

1. Experiment Set-up

(1) Calibration

Voltage #1 : 1.60 V
Voltage #2 : -1.59 V
⇒ Used to calibrate Logger Pro probe

2 second/sample + 3000 s total length
100 stir/min

(2) Material

① 0.001 M $AgNO_3$ in 0.01 M NH_4NO_3
② 0.001 M NH_4Cl ⇒ muddy (milky)
③ 0.1 M $CuSO_4$
④ 0.01 M KI ⇒ green ⇒ milky
⑤ 2.0 M NH_3/NH_4OH in 0.01 M NH_4NO_3 ⇒
clean again

(3) Important Operations

① Electrode

* Clean silver wire in hot Na_2CO_3 solution with aluminium foil
& Careful place copper wire above salt bridge. but not in touch with it

② Salt Bridge

* Prepared already, filled with $CuSO_4$ solution in the electrochemical cell

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Figure 11: Raw Lab Notes